noc25_ee134

Mock Exam 3 | Course ID: noc25_ee134

Q1. In the context of EV charging, what does `EVSE` stand for?

A) Electric Vehicle Safety Equipment
B) Electric Vehicle Supply Equipment
C) Energy Voltage Standardization Entity
D) Electronic Vehicle Service Engine

Source: -GZyYmm0B2M | Difficulty: Easy

Q2. What is the primary safety function of an isolated DC-DC converter in an EV charger?

A) To correct the power factor
B) To provide galvanic isolation between the grid and the battery
C) To reduce the size of the transformer
D) To increase the switching frequency

Source: TlKIeTe3Fy0 | Difficulty: Easy

Q3. What is Zero Voltage Switching (ZVS)?

A) A technique where the current through a switch is zero before it turns on.
B) A technique where the voltage across a switch is zero before it turns on.
C) A technique where the switching frequency is reduced to zero.
D) A technique where the dead time is set to zero.

Source: -kxjMKhO51M | Difficulty: Easy

Q4. What is the primary purpose of the Open Charge Point Protocol (OCPP)?

A) To define the physical dimensions of charging connectors.
B) To ensure interoperability between EVSEs and Central Management Systems (CMS).
C) To regulate the maximum power output of a charging station.
D) To manage the battery's state of health during charging.

Source: UAKL2GPPj0U | Difficulty: Easy

Q5. In Pulse Width Modulation (PWM), what is the 'modulation index (m)'?

A) The ratio of the carrier frequency to the modulating frequency.
B) The ratio of the DC link voltage to the output voltage.
C) The ratio of the peak amplitude of the modulating signal to the peak amplitude of the carrier signal.
D) The total on-time of a switch in one cycle.

Source: VP2T2QgDMws | Difficulty: Easy

Q6. What is the main advantage of a bridgeless PFC converter over a conventional boost PFC converter?

A) It operates at a lower switching frequency.
B) It requires a larger inductor.
C) It has lower conduction losses due to fewer conducting devices.
D) It eliminates the need for an output capacitor.

Source: X4acp5BiIFs | Difficulty: Easy

Q7. What is the defining characteristic of Discontinuous Conduction Mode (DCM) in a PFC converter?

A) The output voltage ripple is zero.

B) The inductor current falls to zero during each switching cycle.
C) The switching frequency is variable.
D) The power factor is always less than 0.9.

Source: FBut4yhA4q0 | Difficulty: Easy

Q8. What type of communication is exclusively used for data exchange between the EV and EVSE in the BEVC-DC001 standard?

A) Power Line Communication (PLC)
B) Pulse Width Modulation (PWM)
C) Controller Area Network (CAN) communication
D) Wi-Fi communication

Source: kd8MFLt2h9k | Difficulty: Easy

Q9. What is the primary architectural difference between an AC EV charger and a DC EV charger?

A) The location of the Battery Management System (BMS).
B) The type of communication protocol used.
C) The location of the main power conversion unit (AC-DC).
D) The physical connector type used.

Source: a-fvKWBRT_g | Difficulty: Easy

Q10. What does the 'Peak Repetitive Reverse Voltage (VRRM)' rating of a diode signify?

A) The maximum forward voltage it can handle.
B) The maximum average current it can conduct.
C) The maximum instantaneous reverse voltage it can withstand repeatedly.
D) The energy it can absorb during a short pulse.

Source: D3_2zrv4rkY | Difficulty: Easy

Q11. For a single-phase full-bridge rectifier with a purely resistive load, what is the characteristic of the current drawn from the AC source?

A) It is a pulsating DC current.
B) It has a high Total Harmonic Distortion (THD).
C) It is sinusoidal and in phase with the input voltage.
D) It flows only during the positive half-cycle of the input voltage.

Source: LInjt3L5-IM | Difficulty: Medium

Q12. What is the primary cause of the large output capacitor size required in single-phase boost PFC converters?

A) The high switching frequency ripple.
B) The need to store energy for the entire charging session.
C) The presence of a second-line harmonic ripple at twice the line frequency.
D) The high peak reverse voltage across the capacitor.

Source: N8Qla_goSZ4 | Difficulty: Medium

Q13. What is a key advantage of the full-bridge DC-DC converter regarding its output filter components?

A) The output ripple frequency is the same as the switching frequency (fsw).
B) It does not require an output inductor.
C) The output ripple frequency is twice the switching frequency (2fsw), allowing for smaller components.
D) The output voltage has no ripple, eliminating the need for a capacitor.

Source: 38D14JN-3wg | Difficulty: Medium

Q14. In AC Type-2 charging, how does the EV signal to the EVSE that it is ready for charging?

A) By sending a high-frequency signal over the PP line.
B) By changing the duty ratio of the PWM signal on the CP line.
C) By changing the voltage level of the PWM signal on the CP line (e.g., from +9V to +6V).
D) By sending a CAN message to the EVSE.
Source: 6-xQQwjzPIc | Difficulty: Medium

Q15. What is the primary cause of zero-crossing distortion (ZCD) in a Totem Pole PFC converter?

A) The use of a large output capacitor.
B) The high switching frequency of the main leg.
C) The elimination of the input diode bridge.
D) Delays and synchronization issues in switch transitions near the voltage zero crossing.
Source: RxJCQteGJQA | Difficulty: Medium

Q16. What is the purpose of the pre-charging stage in the DC CCS2 charging sequence?

A) To verify the compatibility of the communication protocols.
B) To gradually raise the EVSE output voltage to match the EV battery voltage, preventing large inrush currents.
C) To perform an isolation check on the charging cable.
D) To lock the connector in place before high power is applied.
Source: cEaVn-3aSNk | Difficulty: Medium

Q17. Why is the JSON-based implementation of OCPP generally preferred over the SOAP-based implementation?

A) JSON uses XML for data representation, which is more robust.
B) JSON with WebSockets is more lightweight, faster, and scalable.
C) SOAP does not support security protocols.
D) SOAP requires a physical connection for each EVSE, while JSON is wireless.
Source: WUYG4Eko3rU | Difficulty: Medium

Q18. In the context of PWM full-bridge converters, what is the main purpose of introducing a 'dead time'?

A) To increase the duty ratio of the switches.
B) To reduce the switching frequency.
C) To allow device output capacitances to charge/discharge for soft switching and prevent shoot-through.
D) To synchronize the leading and lagging legs.
Source: gLUPdxAcUN8 | Difficulty: Medium

Q19. In the DQ reference frame model of a three-phase AC-DC converter, what must be done to achieve true independent control of the d-axis and q-axis currents?

A) The switching frequency must be increased.
B) The cross-coupling terms (e.g., $\omega L*Isq$) must be compensated for.
C) The modulation index must be set to 1.
D) The winding resistance (Rs) must be eliminated.
Source: 3TF3ZUAokeo | Difficulty: Medium

Q20. In a Phase-Shifted Full Bridge (PSFB) converter, how is the output voltage regulated?

A) By varying the turn-on time (duty cycle) of all four switches.
B) By varying the switching frequency.
C) By varying the phase shift between the leading and lagging legs.
D) By changing the transformer turns ratio.
Source: YINqDUxMemQ | Difficulty: Medium

Q21. How does a flyback-based PFC converter operating in Discontinuous Conduction Mode (DCM) achieve unity power factor?

A) By using a complex current multiplier in its control loop.
B) By presenting a resistive input impedance, making the average input current follow the input voltage.
C) By operating at a very low switching frequency.
D) By ensuring the inductor current is always continuous.

Source: YYOndlU0fVE | Difficulty: Medium

Q22. What is a significant advantage of a balanced three-phase AC-DC converter regarding its output capacitor sizing compared to a single-phase converter?

A) The output voltage is always lower, reducing capacitor voltage stress.
B) It requires a capacitor with a higher ESR.
C) The second line frequency harmonic component is absent, allowing for a much smaller capacitor.
D) It can use a non-polarized capacitor.

Source: inlEtBllhIE | Difficulty: Medium

Q23. The RMS current flowing through the DC-link capacitor (`I_C_RMS`) in a three-phase AC-DC converter is calculated by:

A) Taking the average of the instantaneous capacitor current over a line cycle.
B) Squaring the average converter output current and subtracting it from the square of the RMS converter output current, then taking the square root.
C) Multiplying the peak source current by the modulation index.
D) Integrating the converter output current over one switching cycle.

Source: -dgKwI8Qoqs | Difficulty: Hard

Q24. In a Type-2 PWM strategy for a full-bridge converter, why is Zero Voltage Switching (ZVS) turn-on difficult to achieve for the lagging leg switches at light loads?

A) The dead time for the lagging leg is too short.
B) The leading leg switches consume all the available resonant energy.
C) The output inductor is short-circuited and does not contribute energy; only the small leakage inductance energy is available for capacitor discharge.
D) The body diodes of the lagging leg switches have a high forward voltage drop.

Source: XQsOeitdyiE | Difficulty: Hard

Q25. In transformer design, what is the relationship between total power loss (`P_total`), core loss (`P_core`), copper loss (`P_CU_total`), and peak AC flux density (`ΔB`)?

A) Both `P_core` and `P_CU_total` increase as `ΔB` increases.
B) `P_core` increases with `ΔB`, while `P_CU_total` decreases with `ΔB`, resulting in an optimal `ΔB` for minimum `P_total`.
C) `P_core` decreases with `ΔB`, while `P_CU_total` increases with `ΔB`.
D) Both `P_core` and `P_CU_total` are independent of `ΔB`.

Source: WwzCJVUrYNY | Difficulty: Hard

Q26. During the SLAC procedure in CCS2 DC charging, what is the purpose of the 'BCB toggle'?

A) To exchange the Network Membership Key (NMK) between the EV and EVSE.
B) To measure the initial signal attenuation across all frequency groups.
C) To validate the physical connection by observing a low-frequency voltage change on the CP line when attenuation results are ambiguous.
D) To assign an IPv6 address to the EV's communication controller.

Source: eVsfrIYzqdg | Difficulty: Hard

Q27. What is a key design criterion for the inner current loop controller (`Cc(s)`) in a single-phase Boost PFC converter?

A) It must have a very low gain at DC (s=0).
B) Its bandwidth should be higher than the switching frequency.
C) It must have a very high gain at twice the line frequency (e.g., 100 Hz) to accurately track the current reference.
D) It should be designed to be slower than the outer voltage loop.

Source: kGh_kDgqomA | Difficulty: Hard

Q28. In the closed-loop control design for a three-phase AC-DC converter, how is the proportional gain (KPC) of the inner current loop PI controller typically derived?

A) By setting it equal to the integral gain (KIC).
B) By using pole-zero cancellation and comparing the resulting transfer function to a standard second-order system to achieve a desired damping factor.
C) By measuring the phase margin of the open-loop system directly.
D) By ensuring the controller's bandwidth is exactly half the switching frequency.

Source: KmgcXbLuwno | Difficulty: Hard

Q29. The core selection for a filter inductor is governed by an inequality relating core geometry to specified electrical parameters. Which geometric property combination is used on the left-hand side of this inequality?

A) AC * WA / MLT
B) AC^2 * WA / MLT
C) AC * WA^2 / LC
D) AC * LC / WA

Source: 6X9uRXj0MZQ | Difficulty: Hard

Q30. In the small signal model of a Phase Shifted Full Bridge (PSFB) converter, how are the `DI` and `DV` terms incorporated to account for `D-loss`?

A) They are added as parallel resistors in the equivalent circuit.
B) They are ignored as they are negligible in the small signal approximation.
C) They are used to create additional controlled voltage and current sources that modify the basic buck-derived model.
D) They only affect the steady-state operating point, not the small signal model.

Source: 568-ZYztUIY | Difficulty: Hard

Answer Key

1: B	2: B	3: B	4: B	5: C
6: C	7: B	8: C	9: C	10: C
11: C	12: C	13: C	14: C	15: D
16: B	17: B	18: C	19: B	20: C
21: B	22: C	23: B	24: C	25: B
26: C	27: C	28: B	29: B	30: C